PRODUCT USER MANUAL

For IBI Strong Wave Incidence index IBI_OMI_SEASTATE_swi

Issue: 1.2

Contributors : Álvaro de Pascual,

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MERCATOR OCEAN
INTERNATIONAL
2 avenue de l'Aérodrome de Montaudran, 31400 Toulouse, FRANCE
Tél : +33 5 61 39 38 02 - Fax : +33 5 61 39 38 99
Société civile de droit français au capital de 2 000 000 € - 522 911 577 RCS Toulouse - SIRET 522 911 577 00024

marine.copernicus.eu
mercator-ocean.eu

RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2020/09/11	All	Initial version	Álvaro de Pascual Collar	
1.1	2021/02/23		Changes in dataset IBI_OMI_SEASTATE_swi_anomaly	Álvaro de Pascual Collar	C. Derval
1.2	2025/09/15	All	Review of the document, including updates to the Copernicus product identifiers and the climatological reference period (1980–2024).	Álvaro de Pascual Collar	Copernicus Marine Product Management

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GLOSSARY AND ABBREVIATIONS

MOI	Mercator Ocean International
NetCDF	Network Common Data Form
OMI	Ocean Monitoring Indicator
PUM	Product User Manual
QUID	Quality Information Document
SWI	Strong Wave Incidence
IBI-WAV-MYP	CMEMS product IBI_MULTIYEAR_WAV_005_006 DOI: https://doi.org/10.48670/moi-00030
IBI-WAV-NRT	CMEMS product IBI_ANALYSISFORECAST_WAV_005_005 DOI: https://doi.org/10.48670/moi-00025
SWI	Significant Wave Height

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document is the user manual for the CMEMS Ocean Monitoring Indicator (OMI) IBI_OMI_SEASTATE_swi. The Strong Wave Incidence (SWI) index provides information on the occurrence of stormy wave events in four monitoring regions within the IBI domain. Stormy wave events are defined as instances when the significant wave height (SWH) exceeds the 90th percentile.

The selected monitoring regions are intended to provide a summarized overview of the diverse climatic conditions across the IBI regional domain (Figure 1):

- The **WAV1 region** monitors the influence of stormy conditions along the western coast of the Iberian Peninsula.
- The **WAV2 region** focuses on the variability of stormy conditions in the Bay of Biscay.
- The **WAV3 region** covers the northern half of the IBI domain, which is strongly affected by storms transported by the subpolar front.
- The **WAV4 region** assesses the impact of marine storms along the northeastern African coast, including the Gulf of Cádiz, Canary Islands and Madeira.

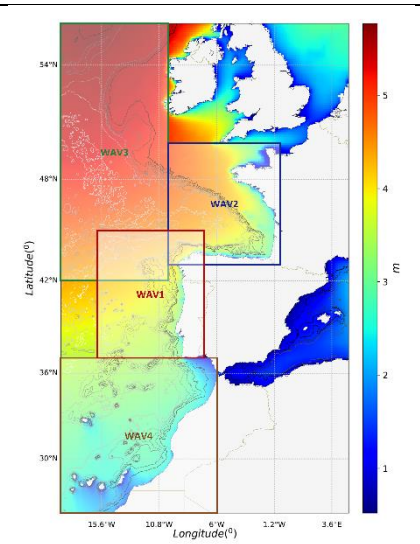

Figure 1: *Mean monthly 90th percentile of significant wave height, computed on an hourly basis from the CMEMS product IBI_MULTIYEAR_WAV_005_006 (climatological reference period 1980–2025). The red, blue, green, and brown squared areas indicate the four monitoring regions defined in the product (WAV1, WAV2, WAV3, and WAV4, respectively).*

The IBI_OMI_SEASTATE_swi product is organized into two datasets.

The anomalies dataset (IBI_OMI_SEASTATE_swi_anomaly) contains the monthly time series of the SWI index, averaged over each monitoring region.

Monitoring Region	Latitude (min/max)	Longitude (min/max)
WAV1	37ºN / 45ºN	16ºW / 7ºW
WAV2	43ºN / 50ºN	10ºW / 1ºE
WAV3	42ºN / 56ºN	19ºW / 10ºW
WAV4	26ºN / 37ºN	19ºW / 6ºW

The mean percentile dataset (IBI_OMI_SEASTATE_swi_mean_percentile) provides the average 90th percentile of significant wave height, computed from hourly data of the IBI-WAV-MYP product.

Two CMEMS products are used to compute the data:

- **IBI-WAV-MYP:** IBI_MULTIYEAR_WAV_005_006
- **IBI-WAV-NRT:** IBI_ANALYSISFORECAST_WAV_005_005

The product is published on the CMEMS dissemination server after undergoing automatic and delayed-mode quality control. It is available online and distributed through the CMEMS Information System. The downloaded files are in NetCDF format and follow the CF-1.4 convention.

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.
Product IBI_OMI_CURRENTS_cui is described in the Quality Information Document (QUID) CMEMS-IBI-QUID-IBI_OMI_SEASTATE_swi.pdf. See also News flash. More detailed information can be obtained from http://marine.copernicus.eu/services-portfolio/contact-us/

b) History of Changes

December 2020	Publication of IBI_OMI_SEASTATE_swi product in monitoring regions WAV1, WAV2, WAV3, and WAV4: • Dataset: IBI_OMI_SEASTATE_swi_anomaly • Dataset: IBI_OMI_SEASTATE_swi_mean_percentile
February 2021	Data from dataset IBI_OMI_SEASTATE_swi_anomaly has been compiled in a single file containing the four wav monitoring regions.
November 2025	The new datasets include the combined temporal coverage of the upstream products: IBI_MULTIYEAR_WAV_005_006 and IBI_ANALYSISFORECAST_WAV_005_005 (from 1980 to April 2025).

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	IBI_OMI_SEASTATE_swi • Dataset: IBI_OMI_SEASTATE_swi_anomaly
Geographical coverage	Iberia-Biscay-Ireland region
Variables	• swi_myp_wav1 • swi_nrt_wav1 • swi_myp_wav2 • swi_nrt_wav2 • swi_myp_wav3 • swi_nrt_wav3 • swi_myp_wav4 • swi_nrt_wav4
Available time series	From 1980 to YEAR-1
Temporal resolution	Monthly means
Horizontal resolution	N/A
Integration depth (for example)	Surface
Format	NetCDF4

Product Lines	IBI_OMI_SEASTATE_swi • Dataset: IBI_OMI_SEASTATE_swi_mean_percentile
Geographical coverage	Iberia-Biscay-Ireland region
Variables	• swh_percentile90_myp
Available time series	01/1980 to 12/2023
Temporal resolution	Monthly means
Horizontal resolution	1/10 degrees (latitude-longitude)
Integration depth (for example)	Surface
Format	NetCDF4

b) Production System Description

The SWI index is defined as the difference between the climatological frequency of exceedance (F_{clim}) and the observed frequency of exceedance (F_{obs}) of the threshold defined by the 90th percentile (Th_{P90}) of SWH, computed on a monthly basis from hourly data of the IBI-WAV-MYP product:

$$SWI = F_{obs}(SWH > Th_{P90}) - F_{clim}(SWH > Th_{P90})$$

Since the SWI is defined as the difference between a climatological mean and an observed value, it can be interpreted as an anomaly. The index represents the percentage by which stormy conditions occur above or below the climatological average. Accordingly, positive/negative values indicate the percentage of hourly data exceeding the threshold above/below the climatological mean, respectively.

The mean percentile is derived from hourly data of the IBI-WAV-MYP product. It is computed on a monthly basis over the full time period (1980–2024) and subsequently averaged.

c) Dataset details

IBI_OMI_SEASTATE_swi_anomalies contains the 1D monthly data of SWI index averaged in the corresponding monitoring region
swi_myp_wav1 [percentage] Anomaly of the percentage of occurrence of stormy conditions above the climatic average computed with IBI-WAV-MYP product in region WAV1. • Long name: Percentage of high significant wave height values above the climatic average • Standard name: wave_significant_height_anomaly **swi_nrt_wav1** [percentage] Anomaly of the percentage of occurrence of stormy conditions above the climatic average computed with IBI-WAV-NRT product in region WAV1. • Long name: Percentage of high significant wave height values above the climatic average • Standard name: wave_significant_height_anomaly **swi_myp_wav2** [percentage] Anomaly of the percentage of occurrence of stormy conditions above the climatic average computed with IBI-WAV-MYP product in region WAV2. • Long name: Percentage of high significant wave height values above the climatic average

- Standard name: wave_significant_height_anomaly

swi_nrt_wav2 [percentage]

Anomaly of the percentage of occurrence of stormy conditions above the climatic average computed with IBI-WAV-NRT product in region WAV2.

- Long name: Percentage of high significant wave height values above the climatic average
- Standard name: wave_significant_height_anomaly

swi_myp_wav3 [percentage]

Anomaly of the percentage of occurrence of stormy conditions above the climatic average computed with IBI-WAV-MYP product in region WAV3.

- Long name: Percentage of high significant wave height values above the climatic average
- Standard name: wave_significant_height_anomaly

swi_myp_wav4 [percentage]

Anomaly of the percentage of occurrence of stormy conditions above the climatic average computed with IBI-WAV-MYP product in region WAV4.

- Long name: Percentage of high significant wave height values above the climatic average
- Standard name: wave_significant_height_anomaly

swi_nrt_wav4 [percentage]

Anomaly of the percentage of occurrence of stormy conditions above the climatic average computed with IBI-WAV-NRT product in region WAV4.

- Long name: Percentage of high significant wave height values above the climatic average
- Standard name: wave_significant_height_anomaly

IBI_OMI_SEASTATE_swi_mean_percentile contains the 2D averaged 90th percentile of significant wave height computed from hourly data:

swh_percentile90_myp [metres]

Averaged 90th percentile of significant wave height computed from hourly data of IBI-WAV-MYP product.

- Long name: Averaged 99th percentile of significant wave height
- Standard name: sea_surface_wave_significant_height

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

5) REFERENCES

Quality Information Document

https://documentation.marine.copernicus.eu/QUID/CMEMS-IBI-QUID-IBI_OMI_SEASTATE_swi.pdf

Pascual A., Levier B., Aznar R., Toledano C., García-Valdecasas JM., García M., Sotillo M., Aouf L., Álvarez E. (2020) Monitoring of wave sea state in the Iberia-Biscay-Ireland regional seas. In von Scuckmann et al. (2020) Copernicus Marine Service Ocean State Report, Issue 4, Journal of Operational Oceanography, 13:sup1, S1-S172, DOI: 10.1080/1755876X.2020.1785097